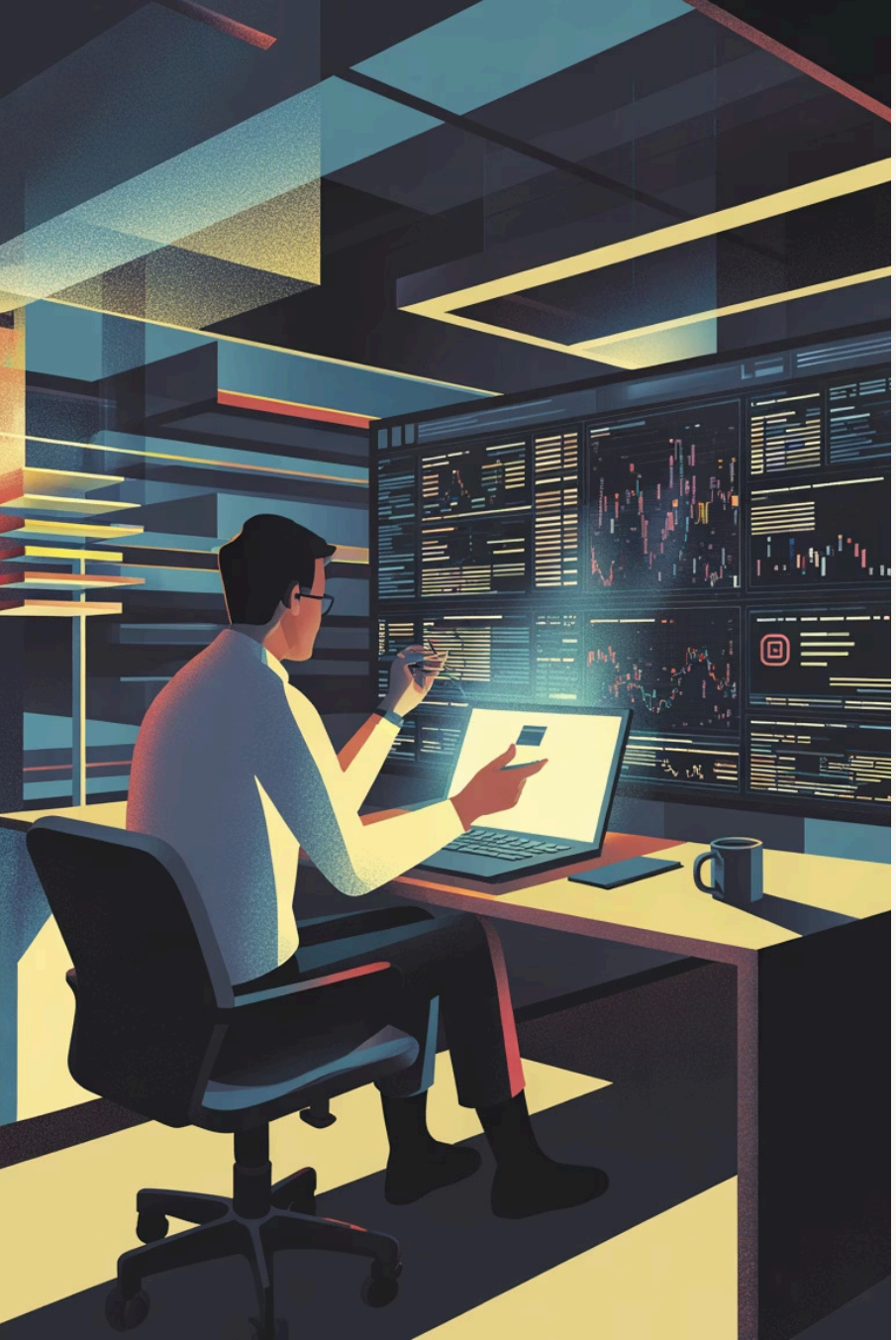

Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** across product categories to guide strategic business decisions.

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Dataset at a Glance

3,900

Purchases

18

Features

$59.76

Avg. Spend

3.75

Avg. Rating

Demographics

Age, Gender, Location, Subscription

Purchase Details

Category, Amount, Season, Size, Color

Behavior

Discounts, Frequency, Ratings, Shipping

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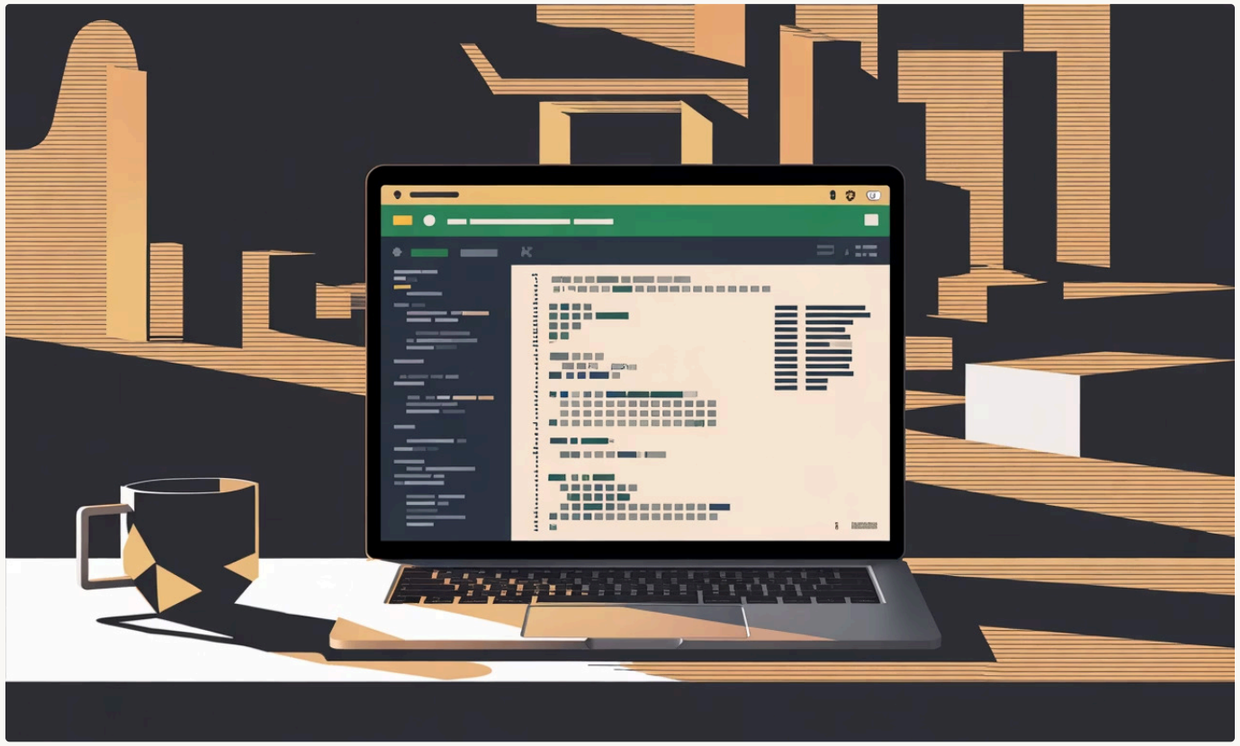
Data Preparation in Python



01

Load & Explore

Imported with `pandas`; used `df.info()` and `.describe()`

02

Clean & Standardize

Imputed missing Review Ratings; renamed columns to snake_case

03

Feature Engineering

Created `age_group` and `purchase_frequency_days` columns

04

Load to PostgreSQL

Connected Python to PostgreSQL; loaded cleaned DataFrame for SQL analysis

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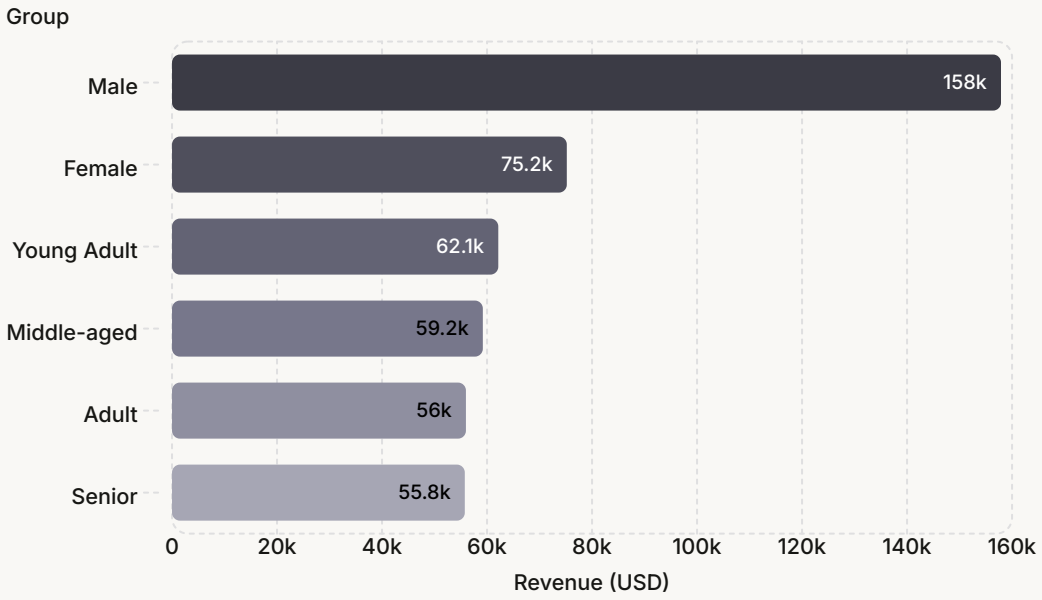
Revenue Insights



Key Findings

- Male customers generate **2× more revenue** than female
- Young Adults are the **top revenue age group** at $62,143
- Revenue is relatively balanced across Adult and Senior segments

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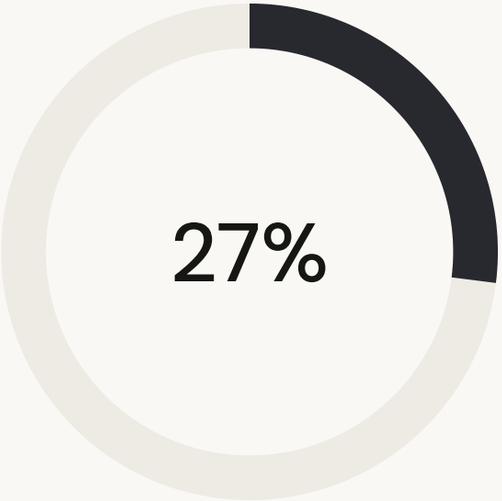
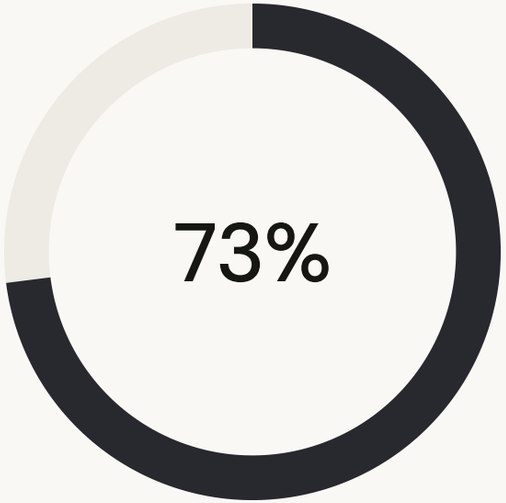
Subscription & Discount Behavior



Subscribers

1,053 customers



Non-Subscribers

2,847 customers

Spend Comparison

Segment	Avg. Spend	Total Revenue
Subscribers	$59.49	$62,645
Non-Subscribers	$59.87	$170,436

839 discount users still spent **above average** — discounts drive volume without sacrificing quality buyers.

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Top Products & Categories

Accessories

Jewelry · Sunglasses · Belt

Clothing

Blouse · Pants · Shirt

Footwear

Sandals · Shoes · Sneakers

Outerwear

Jacket · Coat

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Ratings & Discount Dependency

Top 5 Products by Rating

Product	Avg. Rating
Gloves	3.86 ⭐
Sandals	3.84 ⭐
Boots	3.82 ⭐
Hat	3.80 ⭐
Skirt	3.78 ⭐

Most Discount-Dependent Products

Product	Discount Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

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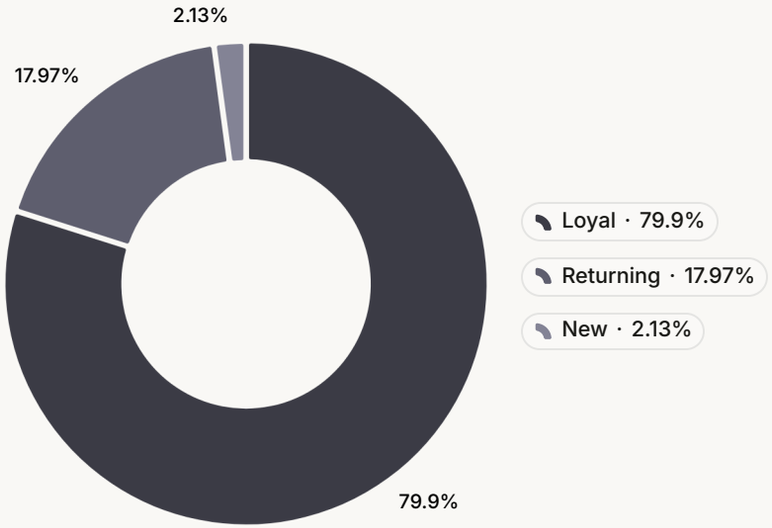
Customer Segmentation



Segment Breakdown

Loyal — 3,116

>5 previous purchases; most likely to subscribe

Returning — 701

Repeat buyers with growth potential

New — 83

Acquisition opportunity

Repeat buyers (>5 purchases): **958 subscribers** vs. 2,518 non-subscribers

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Power BI Dashboard

Interactive Filters

- Subscription Status
- Gender
- Category
- Shipping Type

Visualizations

- Revenue & Sales by Category
- Revenue & Sales by Age Group
- Subscription donut (27% Yes / 73% No)

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Business Recommendations

Boost Subscriptions

Promote exclusive benefits to convert the 73% non-subscribers

Loyalty Programs

Reward repeat buyers to grow the Loyal segment

Review Discount Policy

Balance sales boosts with margin control

Targeted Marketing

Focus on high-revenue age groups and express-shipping users

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